

DS&AI

Machine Learning

DecisionTree

Q1 Consider a binary classification problem with the following dataset

Sample	Feature 1	Feature 2	Class label
A	2	3	0
B	1	5	1
C	4	2	1
D	3	1	0
E	5	4	0
F	2	2	1
G	3	3	1

Calculate the Gini impurity for the entire dataset

- (A) 0.514 (B) 0.486
(C) 0.353 (D) 0.65

Q2 We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

What is the entropy $H(\text{Passed})$?

- (A) 0.46 (B) 0.63
(C) 0.92 (D) 0.23

Q3 In a binary classification problem, a classifier correctly predicts 90 instances as positive, incorrectly predicts 15 instances as positive when they are negative, correctly predicts 90 instances as negative, and incorrectly predicts

10 instances as negative when they are positive. What is the accuracy of the classifier?

- (A) 80 (B) 83
(C) 88 (D) 93

Q4 Which of the following statements is/are true about Gini Impurity in Decision trees?

- (A) Gini index is a measure of impurity or purity used while creating a decision tree in the CART
(B) Gini Impurity is the measurement of changes in entropy after the segmentation of a dataset based on an attribute
(C) An attribute with a high Gini index should be preferred as compared to the low Gini index
(D) It only creates binary splits, and the CART algorithm uses the Gini index to create binary splits.

Q5 Suppose you are working on a binary classification problem, and you decide to use logistic regression to model the relationship between a single feature (X) and the binary outcome (Y). You collect a dataset and fit the logistic regression model:

$$P(Y = 1|X) = \frac{1}{1 + e^{(\beta_0 + \beta_1 X)}}$$

After training, you obtain the following logistic regression coefficients:

$$b_0 = -3.2$$

$$b_1 = 0.8$$

Now, you want to make predictions and evaluate the model's performance

For a given observation with $X=2.5$, calculate the predicted probability of $Y = 1$, and predict the class



- (A) 0.99, class 1 (B) 0.23, class:0
(C) 0.23, class 1 (D) 0.77, class:0

Q6 Consider a dataset with the following distribution of classes for a binary classification problem:

Class A: 40 instances

Class B: 60 instances

After the first split based on a certain feature, the dataset is divided into two subsets:

Subset 1: Class A: 30 instances

Class B: 10 instances

Subset 2: Class A: 10 instances

Class B: 50 instances

Determine the information gain achieved by the split

- (A) 0.787 (B) 0.389
(C) 0.307 (D) 0.26

Q7 Assume that you are given a data set and a neural network model trained on the data set. You are asked to build a decision tree model with the sole purpose of understanding/interpreting the built neural network model. In such a scenario, which among the following measures would you concentrate most on optimising?

(A) Accuracy of the decision tree model on the given data set
(B) F1 measure of the decision tree model on the given data set
(C) Fidelity of the decision tree model, which is the fraction of instances on which the neural network and the decision tree give the same output
(D) Comprehensibility of the decision tree model, measured in terms of the size of the corresponding rule set

Q8 In a decision tree, if a node has 3 possible outcomes with probabilities 0.4, 0.3 and 0.3 what is the entropy? (Rounded up to 2 decimal)

Q9 Consider the following statements:

S₁: Decision Trees are prone to overfitting, especially when they are allowed to grow deep without pruning.

S₂: Cross-Validation provides an unbiased estimate of model performance by using multiple training and validation splits.

Which of the above is true?

- (A) Only S₁ (B) Only S₂
(C) Both S₁ and S₂ (D) None of these

Q10 Consider the following statements:

S₁: A Decision Tree is a generative model.

S₂: Cross-Validation is used to evaluate the performance of a model on unseen data.

Which of the above is true?

- (A) Only S₁ (B) Only S₂
(C) Both S₁ and S₂ (D) None of these

Q11 Which of the following criteria is used in the top-down decision tree approach to evaluate the best split at a node?

- (A) Gini Index
(B) Mean Squared Error (MSE)
(C) Information Gain
(D) Kappa Statistic

Q12 Consider the following statements:

S₁: The depth of a decision tree directly controls its bias and variance.

S₂: Decision trees are naturally resistant to overfitting.

S₃: Entropy in decision trees is a measure of randomness in the data.

S₄: In decision trees, features are chosen for splitting based on their correlation with the output variable.

Which of the above statements is correct?

- (A) Only S₁ and S₂ (B) Only S₂ and S₃
(C) Only S₁ and S₃ (D) Only S₁ and S₄

Q13 Consider a node in a decision tree that contains 80 samples. Out of these, 50 samples



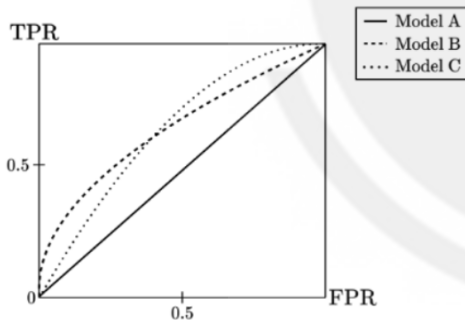
belong to Class X, 20 samples belong to Class Y and 10 samples belong to Class Z. Calculate the Gini Index for this node.

- (A) 0.32 (B) 0.46
(C) 0.53 (D) 0.38

Q14 A decision tree classifier learned from a fixed training set achieves 100% accuracy. Which of the following models trained using the same training set will also achieve 100% accuracy?

- (i) Logistic regressor.
(ii) A polynomial of degree one kernel SVM.
(iii) A linear discriminant function.
(iv) Naïve Bayes classifier.
(A) i
(B) i and ii
(C) All of the above
(D) None of the above

Q15 Suppose we fix the decision threshold for all 3 logistic regression models to be such that we get a FPR of 0.8 when we evaluate our model. In order of most to least preferred, rank each of the models given their ROC curves.



- (A) $C > B > A$
(B) $C > A > B$
(C) $B > C > A$
(D) $B > A > C$

Q16 Consider training a decision tree given a

design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels

$$y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ Let } f_1 \text{ denote feature 1,}$$

corresponding to the first column of X , and let f_2 denote feature 2, corresponding to the second column. Which of the following splits at the root node gives the highest information gain?

- (A) $f_1 > 2$ (B) $f_2 > 3$
(C) $f_1 > 4$ (D) $f_2 > 6$

Q17 Which of the following is true for a decision tree?

- A. A decision tree is an example of a linear classifier.
B. The entropy of a node typically decreases as we go down a decision tree.
C. Entropy is a measure of purity.
D. An attribute with lower mutual information should be preferred to other attributes.
(A) A (B) B
(C) C (D) D

Q18 Consider the dataset, S given below:

Elevation	Road Type	Speed Limit	Speed
Steep	Uneven	Yes	Slow
Steep	Smooth	Yes	Slow
Flat	Uneven	No	Fast
Steep	Smooth	No	Fast

Elevation, Road Type and speed Limit are the features and Speed is the target label that we want to predict.

Find the entropy of the dataset, S as given above:

- A. 0.5
B. 0
C. 1
D. 0.7
(A) A (B) B
(C) C (D) D



Q19 Find the information Gain if the dataset is split at the feature "Elevation":

- A. 1
- B. 0
- C. 0.675
- D. 0.311

(A) A (B) B
(C) C (D) D

Q20 You have generated data from a 3-degree polynomial with some noise. What do you expect of the model that was trained on this data using a 5- degree polynomial as function class?

- A. Low bias, high variance
- B. High bias, low variance
- C. Low bias, low variance
- D. High bias, low variance

(A) A (B) B
(C) C (D) D

Q21 Given the following table of observations, calculate the information gain $IG(YX)$ that

would result from learning the value of X

X	Y
Red	True
Green	False
Brown	False
Brown	False

- (A) 1/2
- (B) 1
- (C) 3/2
- (D) None of the above

Q22 When choosing one feature from $X_1, \dots, X_n$ while building a Decision Tree, which of the following criteria is the most appropriate to maximize? (Here, $H()$ means entropy, and $P()$ means probability)

- (A) $P(Y|X_j)$
- (B) $P(Y) - P(Y|X_j)$
- (C) $H(Y) - H(Y|X_j)$
- (D) $H(YX_j)$
- (E) $H(Y) - P(Y)$



Answer Key

Q1 A
Q2 A
Q3 C
Q4 A
Q5 A
Q6 D
Q7 C
Q8 1.5709
Q9 C
Q10 B
Q11 A, C

Q12 C
Q13 C
Q14 D
Q15 A
Q16 C
Q17 B
Q18 C
Q19 D
Q20 A
Q21 B
Q22 C



Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Entire data

0 : 3

1 : 4

$$P_0 = 3/7 \quad P_1 = 4/7$$

$$GI = 1 - P_0^2 - P_1^2 = 0.489$$

Q2 Text Solution:

$$H(P) \quad f : 2 \quad P_F = 1/3$$

$$\frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 3/2 = 0.918$$

Q3 Text Solution:

Actual

	P	N
P	90	15
N	10	90

$$\text{Accuracy} = \frac{T_P + T_N}{\text{Total}} \times 100$$

$$= \frac{90+90}{205} \times 100$$

$$= 87.8\%$$

Q4 Text Solution:

Gini index is a measure of impurity or purity used while creating a decision tree in the CART

Q5 Text Solution:

Simple put value in formula

$$P_1 = 0.99 \quad \text{Class1}$$

$$P_1 = \frac{1}{1+e^{0.8-2.5 \times 3.2}} = 0.999$$

Q6 Text Solution:

$$\text{Parent } P_A = 0.4 \quad P_B = 0.6$$

$$\epsilon = 0.970$$

$$\text{Child : subset 1 } P_A = 3/4 \quad P_B = 1/4$$

$$\epsilon_{S1} = 0.811$$

$$\text{Subset 2 } P_A = 1/6 \quad P_B = 5/6$$

$$\epsilon_c = \frac{60 \times 0.650 + 40 \times 0.811}{100} = 0.7144$$

$$IG = \epsilon_P - \epsilon_c = 0.26$$

Q7 Text Solution:

Here we want DT to mimic O1 to perform similar to NN

So we want similar performance

Q8 Text Solution:

$$Ex =$$

$$0.4 \log_2 \frac{1}{0.4} + 0.3 \log_2 \frac{1}{0.3} + 0.3 \log_2 \frac{1}{0.3} = 1.5709$$

Q9 Text Solution:

- **S1 (True):** Decision Trees are highly prone to overfitting, especially when they are allowed to grow deep without pruning. This happens because deep trees tend to capture noise in the training data rather than generalizing well to unseen data. Pruning or limiting the depth of the tree helps mitigate this issue.
- **S2 (True):** Cross-Validation provides an unbiased estimate of model performance by splitting the dataset multiple times into training and validation sets. This technique ensures that the model is evaluated on different subsets of data, reducing bias and variance in performance estimation.

both statements are correct

Q10 Text Solution:

S1 (False): A Decision Tree is a **discriminative model**, not a generative model.

- **Generative models** learn the joint probability distribution $P(X, Y)$



and can generate new data points (e.g., Naïve Bayes, Gaussian Mixture Models).

- **Discriminative models** learn the decision boundary by modeling $P(Y|X)$, which is what Decision Trees do.
- **S2 (True):** Cross-Validation is used to evaluate a model's performance on unseen data by splitting the dataset into multiple training and validation subsets. This helps assess how well the model generalizes to new data.

S2 is correct

Q11 Text Solution:

In a top-down **Decision Tree approach**, the best split at a node is determined using different criteria depending on whether the problem is **classification** or **regression**:

- **For Classification:**
 - **Gini Index** (Option A): Measures the impurity of a node; lower values indicate a purer node. Used in **CART (Classification and Regression Trees)**.
 - **Information Gain** (Option C): Measures the reduction in entropy before and after a split; used in **ID3, C4.5, and C5.0 algorithms**.
- **For Regression:**
 - **Mean Squared Error (MSE)** (Option B) is used in regression trees, but the question specifically asks for a **top-down decision tree approach** (which usually refers to classification trees).
- **Kappa Statistic** (Option D) is used to measure the agreement between classifiers and is not a splitting criterion in decision trees.

A) Gini Index and C) Information Gain

Q12 Text Solution:

S1

- The depth of a decision tree **directly controls its bias and variance**.
- A **shallow tree (low depth)** has **high bias** and may underfit.
- A **deep tree (high depth)** has **low bias but high variance**, leading to overfitting.

S3

- **Entropy** is a measure of **randomness or impurity** in data.
- A higher entropy means the data is more randomly distributed (less pure).
- Decision trees use **Information Gain** (based on entropy) to determine the best splits.

Q13 Text Solution:

$$\begin{aligned} \text{GI} &= 1 - (P_x^2 + P_y^2 + P_z^2) \\ &= 1 - (5/8)^2 - (2/8)^2 - (1/8)^2 \\ &= 0.531 \end{aligned}$$

Q14 Text Solution:

Decision tree = 100% accuracy

Overfit

So we want to search algo that do overfitting

Q15 Text Solution:

So IFPR = 0.8

So, We select algo that give higher TPR model
 $C > B > A$

Q16 Text Solution:

a) $f_1 > 2 \quad 697 \quad 1, 1, 0$

$2 \quad 0$

So, we need min entropy of child

b) $f_2 > 3 \quad 23 \quad 1, 0$



6, 7 1, 0

c) $f_1 > 4$ 6, 9 = 1, 1

2, 4 = 0, 0

So, we need min entropy of child

d) $f_2 > 6$ 7 0

2, 3, 6 1, 1, 0

Q17 Text Solution:

"Information gain" we choose attribute with max MI or IG

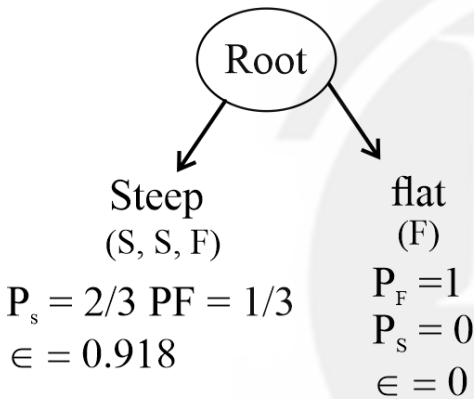
Q18 Text Solution:

Level $H(S)$ $P_S = 1/2$

$P_F = 1/2$

$\epsilon = 1$

Q19 Text Solution:



$$\epsilon_c = \frac{0.918 \times 3 + 1 \times 0}{4}$$

$$= .6888$$

$$IG = \epsilon_P - \epsilon_c = 0.311$$

Q20 Text Solution:

The **true data-generating function** is a **3-degree polynomial** with noise.

The **model is a 5-degree polynomial**, which is **more complex than necessary**.

Q21 Text Solution:

Information Gain wrt

R_{wt}

$$P_F = 3/4$$

$$\epsilon_R = 0.811 \quad P_T = 1/4 \xrightarrow{B} F$$

$\downarrow R \quad \downarrow G$

T F

$$IG = \epsilon_R - \epsilon_c = 0.811 - 0 = 0.811$$

100% homogeneous

$$\epsilon = 0$$

Q22 Text Solution:

$$\max IG = H(Y) - H(Y|x)$$

y is label

x is dimension



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